

The Productivity Gap

Why GenAI's 80% Exposure Yields a 1.56% Macro Forecast

Audit synthesis

Eloundou et al. (2024), *Science* 384, 1306 | Noy & Zhang (2023), *Science* 381, 187
Acemoglu (2025), *Economic Policy* 40(121) | Agrawal, Gans & Goldfarb (2025), NBER ETAI
Demirci, Hannane & Zhu (2025), WP | Goldberg & Lam (2026), Stanford GSB / UCLA

80% — 1.56%

Eight in ten US workers have $\geq 10\%$ of their tasks LLM-exposed.

The 10-year GDP gain is bounded at 1.56%.

Both numbers are right.

Eloundou et al. (2024), Fig. 2; Acemoglu (2025), Table 1, p. 51.

Three pieces of math separate exposure from growth

Layer	Question	Number
Exposure	What share of tasks could AI touch?	80% workers, 4.6% of GDP
Lab gain	How fast/better does AI make a task?	-0.8 SD time, +0.4 SD grade
Macro envelope	Hulten: $d \ln TFP = \bar{\pi} \times s$	$\leq 0.66\%$ TFP / 1.56% GDP
Market reality	Where does the math actually hit?	Narrow sectors, sharp falls
Theory	Who gets displaced first?	Routine, where $\eta_{AI} \leq 1/q^2$

Each row pins down one paper. Read together, they explain why the lab is loud and the macro is quiet.

The headline numbers

Big exposure, big lab gains, small macro envelope.

LLM exposure rises with wages — reversing prior automation's bias

Job zone	Median income	Mean exposure	Share >50%
1 (no prep)	\$30,230	0.06	0.0%
2	\$38,215	0.16	6.1%
3	\$54,815	0.26	10.6%
4 (bachelor's)	\$77,345	0.47	34.5%
5 (master's+)	\$81,980	0.43	26.5%

34 occupations have zero exposed tasks — all manual: roofers, dishwashers, slaughterers, pile-driver operators. Only **1.86%** of all 19,265 tasks are fully automatable (T4).

Eloundou et al. (2024), Tables 1, 5, 8–9; metric is $E1 + 0.5 \cdot E2$.

In the lab, ChatGPT cuts time 0.8 SD and raises grades 0.4 SD

Outcome (Δ post–pre)	ITT (OLS)	TOT (IV)	N
Time on task (minutes)	–10.84***	–16.51***	346
Grade (1–7 scale)	+0.65***	+0.95***	1,135
Job satisfaction (1–10)	+1.16***	+1.93***	449

Drafting time **halves**; editing time **doubles**. ChatGPT operates as a *drafting substitute*, not an ideation aid — 58 of 173 users submitted output **unedited**.

Noy & Zhang (2023), pre-registered RCT, $N=450$ Prolific, 5 white-collar tasks; SI Tables S.1–S.2.

Hulten arithmetic bounds 10-year TFP gain at 0.66%

Component	Value	Source
Wage-bill share exposed	20%	Eloundou (2024)
× cost-effectively automatable	23%	Svanberg et al. (2024)
GDP share affected	= 4.6%	—
Avg. overall cost savings $\bar{\pi}$	14.4%	Noy/Brynjolfsson
$d \ln \text{TFP} = s \times \bar{\pi}$	= 0.66%	p. 41
GDP gain (capital response, 10 yrs)	≤ 1.56%	Table 1, p. 51

An order of magnitude below Goldman's 7% and McKinsey's 1.5–3.4 pp/yr — those forecasts cannot be reconciled with task-level estimates inside Hulten.

Acemoglu (2025), eq. 14 (p. 28); Table 1 cols. 3 & 7 (p. 51).

Where the math hits

The macro is small. The dislocations are not.

ChatGPT cut writing-prone freelance posts 21% within 8 months

Cluster	DiD coef. (SE)	% change	Shock
Writing	−0.362*** (0.054)	−30.4%	ChatGPT
Software / web	−0.231** (0.054)	−20.6%	ChatGPT
Engineering	−0.110 (0.058)	−10.4%	ChatGPT
Graphic design	−0.204** (0.048)	−18.5%	Image AI
3D modeling	−0.169** (0.048)	−15.6%	Image AI

Surviving employers post **higher budgets** (+5.7%), **more complex jobs** (+2.2%), **more bids** (+8.6%) — the routine layer is skimmed off, not the complex one.

Demirci et al. (2025), Tables 2–3, D1; $N = 1,218,463$ posts.

On Adobe Stock, GenAI more than doubled supply **and** cut non-GenAI artists 29%

Outcome (Post 7+ mth × Treated)		All images	Non-GenAI
Production (Poisson APE)		+135.9%***	-14.7%***
Active artists		+47.1%***	-29.0%***
Sales (Poisson APE)		+82.0%***	-28.2%***

Expansion and crowd-out coexist. Non-GenAI artists differentiate — spreading in embedding space where GenAI clusters do not yet reach. Embedding quality of remaining non-GenAI rises +1.8%***.

Goldberg & Lam (2026), Tables 2–4; 1,719 markets, $N = 55,008$.

Theory closes the gap

Routine is fully displaced; geniuses retreat to the frontier.

Routine workers are fully displaced when AI's curvature $\eta_{AI} \leq 1/q^2$

Regime	Routine	Human genius	AI genius
Pre-AI	Interior $[x_0 - q, x_0 + q]$	By $AA(x)$ peak	—
Short-run AI	Unchanged	Pushed outward	Closer + unanswered
Long-run, $\eta_{AI} \leq 1/q^2$	Fully displaced	Frontier only	Routine band
Long-run, $\eta_{AI} > 1/q^2$	Narrow edge band	Frontier	Interior

The displacement condition is a **geometry inequality**, not a productivity threshold: routine's linear value loses wherever AI's quadratic curvature is shallow enough.

Agrawal, Gans & Goldfarb (2025), Propositions 1–3.

A skeptic says: the macro estimate is too low, the J-curve is coming

The objection. The 1.56% ceiling counts only existing tasks at today's $\sim 14\%$ cost savings. Goldman's 7% and McKinsey's 1.5–3.4 pp/yr could be right if AI generates **new tasks** the rubric does not see.

The response.

- **Adoption is a headwind, not a tailwind.** Only 1.5% of US firms had invested in AI by 2019 — 10-year cost reductions are likely overstated.
- **New tasks have a sign problem.** Acemoglu's bad-task calibration: AI-enabled social media + IT-security $\approx 2\%$ of GDP, ratio of harm-to-benefit $-19/53$. Apparent $+2\%$ GDP coexists with -0.72% welfare.
- **Micro converges with macro.** Demirci's 21% drop and Goldberg's 29% artist exit are *narrow* — exactly what Hulten predicts when 4.6% of GDP is exposed at $\bar{\pi} \approx 14\%$.

Bigger gains require new pro-worker tasks — not better chatbots

Layer	What current AI delivers	What would lift the ceiling
Exposure	Broad, progressive (Eloundou)	New-task exposure measures
Lab gain	Drafting substitution (Noy)	Expertise provision: nurses, electricians
Macro	0.66% TFP, 1.56% GDP	New pro-worker tasks Hulten ignores
Market	Routine skimmed (Demirci, Goldberg)	Differentiation niches survive
Theory	Humans $\rightarrow$ frontier (Agrawal)	$\eta_{AI} > 1/q^2$: complement, not substitute

The Hulten ceiling is not a law of nature — it is **what current AI is built to do**.

The macro is small.

The dislocations are **not**.